

When Corruption Becomes a Label: Diagnosing and Mitigating Class-Corruption Shortcuts in Model-Heterogeneous Federated Learning

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Abstract

Model-heterogeneous federated learning (HFL) allows clients with different architectures to collaborate, but corruption-robust HFL usually treats corruption as nuisance variation. We study **class-corruption entanglement (CLE)**, a controlled setting in which client-specific associations between task classes and corruption operators can make corruption a directional class cue. We evaluate the same semantic source under multiple operators and introduce **Directional Shortcut Alignment (DSA)**, which measures whether probability mass moves toward the classes associated with the applied operator during training. Under explicit assumptions about semantic preservation, pre-registered bindings, and evaluation isolation, DSA measures an operator-response contrast along the training binding; a proxy non-identifiability result shows why pseudo-environment improvements alone cannot certify reduced CLE dependence. A seed-0 matched HFL-versus-Local factorial finds a predominantly local-first formation pattern, motivating a local intervention. We combine a taxonomy-assisted coarse **Public Environment Witness (PEW)** with **Balanced Environment Risk (BER)** to compress class-conditional pseudo-environment support imbalance without changing communication. On a held-out binding map over three matched training seeds, PEW+BER attains lower pooled DSA and higher pooled task utility than ERM, CVaR-DRO, and PEW+GroupDRO. Seed-0 experiments also reproduce DSA suppression on another binding map and under a second protocol-matched HFL communication base, where the utility gate fails. The method is taxonomy-assisted, and shortcut suppression does not imply uniform utility gains.

1 Introduction

Federated learning lets multiple clients train collaboratively without sharing their private training data. In model-heterogeneous federated learning (HFL), clients may also use different architectures, which rules out direct parameter averaging and motivates knowledge transfer through predictions, prototypes, or public data. FedDF distills ensemble predictions over unlabeled data [10]; RHFL addresses noisy labels under model heterogeneity [4]; and AugHFL and its TPAMI extension RAHFL study heterogeneous collaboration under common data

corruption [5, 6]. These methods treat corruption mainly as label-preserving nuisance variation, consistent with standard corruption benchmarks [2, 8].

We study a different failure mode. Suppose a client sees most images of one class with blur and most images of another class with noise, while another client has a different class-corruption association. Corruption can then become predictive of the task label. We call this controlled setting **class-corruption entanglement in HFL (CLE-HFL)**. The central question is not only whether corruption reduces accuracy, but whether changing the corruption operator for the same semantic source moves predictions toward the classes associated with that operator during training. Ordinary corrupted accuracy cannot separate this directional shortcut from image degradation, label imbalance, model capacity, or non-IID optimization.

We therefore use a source-paired operator diagnostic. AugMix-style Jensen-Shannon consistency can align augmentations around one corrupted view while leaving cross-operator class routing unchanged [2]. A pseudo-environment score can also improve even when the model still follows the true class-corruption binding. We define **Directional Shortcut Alignment (DSA)** as the change in probability mass toward the classes pre-registered as bound to the applied operator, using the same semantic source across operator interventions. DSA measures response along the training binding under assumptions of semantic preservation, valid operator intervention, pre-registered bindings, source pairing, and evaluation isolation. We also show that, with an unconstrained proxy error channel, proxy statistics alone do not identify true CLE dependence; final mitigation claims therefore require a target-aligned behavioral endpoint.

The paired diagnostic also lets us ask where the shortcut forms. In a seed-0 matched HFL-versus-Local factorial, the strong-CLE contrast is similar with and without communication, while the additional communication contrast is much smaller. We describe this pattern as **predominantly local-first** and target the client objective rather than redesigning communication. Our intervention uses a **coarse family-level Public Environment Witness (PEW)** trained on public synthetic corruptions and **Balanced Environment Risk (BER)**, which compresses support-induced risk-mass differences across pseudo-environments within each task class. PEW is explicitly taxonomy-assisted and does not read private true

corruption operators or families.

We test the intervention along two comparison axes. Axis I compares ERM, a protocol-matched CVaR-DRO control, PEW+GroupDRO with the same PEW grouping information, and PEW+BER on a held-out binding map over training seeds 0/1/2. PEW+BER has the lowest pooled DSA and the highest pooled task utility among these four matched objectives. Shared validation changes only the binding map or the HFL communication base; DSA suppression reproduces in both tests, although the FedDF-fidelity utility gate fails. Axis II is reserved for protocol-matched HFL mechanism-family context and remains pending until Formal results are available. Our contributions are:

1. **CLE-HFL problem formulation.** We formulate client-specific class-corruption bindings in model-heterogeneous FL and separate directional shortcut use from generic corruption difficulty.
2. **Paired diagnosis and identification boundary.** We introduce source-paired operator evaluation and **Directional Shortcut Alignment (DSA)**, characterize its identification and source-level inference, and prove a proxy non-identifiability result that separates proxy improvement from target-aligned CLE mitigation.
3. **Local-versus-communication formation analysis.** A matched HFL-versus-Local factorial shows a predominantly local-first pattern in the evaluated seed-0 setting, with a smaller pooled communication add-on.
4. **Taxonomy-assisted local mitigation and controlled validation.** We combine coarse family-level PEW with BER and evaluate it against matched ERM, tail risk, and group-robustness controls, then test different binding maps and communication bases while reporting utility and taxonomy limits separately.

Prior work already studies federated spurious features, invariant or group-robust learning, environment inference, and counterfactual shortcut evaluation [1, 12, 14, 17–19]. Our scope is their specific intersection with client-dependent class-corruption bindings in model-heterogeneous FL, together with a paired directional diagnostic and local-versus-communication formation analysis.

2 Related Work

2.1 Model-Heterogeneous Federated Learning

Model heterogeneity prevents direct parameter averaging across clients with different architectures. FedDF uses ensemble distillation over unlabeled data [10]. Prototype-based methods communicate class representations: FedProto aggregates client prototypes, whereas FedTGP learns trainable global prototypes with an adaptive-margin contrastive server objective [16, 20]. RHFL addresses noisy labels with confidence-weighted collabora-

tion [4]. AugHFL studies common corruption under model heterogeneity, and RAHFL extends this line with diversity-enhanced local representation learning and asymmetric collaboration [5, 6]. Our setting keeps model heterogeneity but asks whether client-specific class-corruption associations create directional class cues and whether those cues arise mainly during local learning or collaboration.

2.2 Corruption Robustness and Consistency Learning

CIFAR-C and ImageNet-C established common-corruption evaluation across noise, blur, weather, and digital distortions [8]. AugMix combines stochastic augmentations with Jensen-Shannon consistency to improve corruption robustness and uncertainty [2]. CLE-HFL changes the statistical structure: corruption operators are associated with task classes during client training. The diagnostic therefore compares different operators for the same semantic source rather than only enforcing consistency around one corrupted observation.

2.3 Spurious Correlation, Group Robustness, and Environment Inference

Shortcut learning and group distribution shift are well studied [7]. GroupDRO optimizes worst-group risk with known groups [14]; GEORGE recovers hidden subclasses before group-robust training [15]; EIL infers environments without a predefined taxonomy [1]; and JTT up-weights examples misclassified by an initial ERM model [11]. CVaR and related DRO objectives emphasize upper-tail loss [3, 13]. In our setting, PEW supplies a taxonomy-assisted corruption-family proxy, and BER changes the effective risk mass assigned to class-pseudo-environment supports; pseudo-group discovery and group reweighting themselves are established ideas. Our CVaR-DRO arm is a protocol-matched empirical tail-risk control rather than a line-by-line reproduction of either cited formulation.

2.4 Federated Spurious Learning and Counterfactual Diagnosis

Federated spurious features have direct precedents. Personalized federated learning can exploit client-specific spurious features [19]; FedPIN targets shortcut-averse personalized invariant learning [17]; and FedCD studies spurious correlation in federated domain generalization [12]. The current citation audit treats FedCD as an arXiv preprint. Counterfactual shortcut evaluation also appears in other domains, including medical imaging [9, 18]. We specialize this idea to paired interventions that change only the corruption operator for a fixed source, score the response along a pre-registered client binding, and pair that diagnostic with local-versus-federated formation analysis and a local mitigation objective.

3 Problem Setup

3.1 Model-Heterogeneous CLE

We consider clients $k \in \{1, \dots, K\}$, each with a private dataset and classifier f_k ; architectures need not match across clients. The primary controlled setup uses four heterogeneous clients with ResNet10, ResNet12, ShuffleNet, and MobileNetV2 on a CIFAR-10 private task. Client label distributions are non-IID under a Dirichlet partition with concentration $\alpha = 0.5$.

Let Y denote the task class, O a concrete corruption operator, and $F = \pi(O)$ its coarse corruption family. CLE-HFL introduces client-specific class dependence,

$$P_k(O = o \mid Y = c) \neq P_k(O = o),$$

with a pre-registered client-specific binding

$$b_k : \mathcal{C} \rightarrow \mathcal{O}.$$

For $M = |\mathcal{O}_{\text{seen}}|$ seen operators, the training operator is sampled according to

$$P_k(O = o \mid Y = c) = \frac{1 - \gamma_{\text{CLE}}}{M} + \gamma_{\text{CLE}} \mathbf{1}\{o = b_k(c)\}.$$

The control setting uses $\gamma_{\text{CLE}} = 0$, while strong CLE uses $\gamma_{\text{CLE}} = 0.9$, making $b_k(c)$ a strong but non-deterministic dominant operator for class c . The formal paired evaluation uses 1,000 semantic sources, 15 operators, four clients, and corruption severity 3.

CLE-v2 and DSA use operator-level bindings, whereas the deployable intervention remains coarse family-level. PEW predicts a corruption-family proxy, and BER groups private fit samples by task class and that proxy. The training rule never reads private operator metadata. Operator identities and the binding map are reserved for controlled CLE construction, sealed DSA evaluation, and non-deployable Oracle audits.

3.2 Information Separation

Private data are separated by role. The **private fit** subset produces local gradients and BER group counts. A **private audit** subset may support strict communication routing but does not produce local-training gradients. The **final task test** is used only for frozen utility reporting. The **paired operator grid** is used only after training for DSA. Public data support PEW training or pre-existing communication without private-task ground-truth leakage.

During training, the method reads neither private true operator IDs nor true corruption families, evaluation severity, seen/unseen flags, or the paired evaluation binding. The frozen binding is read only after model sealing to compute DSA. This separation keeps the target-aligned diagnostic outside optimization, routing, tuning, and model selection.

4 Paired Diagnosis and Identification

4.1 Directional Shortcut Alignment

For source $z = (x, y)$, let $T_o(x)$ be a semantic-preserving corruption transformation. For client k and operator o , define the classes bound to that operator during training:

$$B_{k,o} = \{c : b_k(c) = o\}.$$

We evaluate only source-operator pairs for which $y \notin B_{k,o}$, avoiding a trivial contribution from the source's true class. Define bound probability mass

$$m_{k,o}(z, o') = \sum_{c \in B_{k,o}} p_k(c \mid T_{o'}(x)).$$

The source-level directional contrast is

$$d_{k,o}(z) = m_{k,o}(z, o) - \frac{1}{|\mathcal{O}| - 1} \sum_{o' \neq o} m_{k,o}(z, o'),$$

and

$$\text{DSA}_{k,o} = \mathbb{E}_z[d_{k,o}(z)].$$

Pooled DSA equally averages valid operator-level values and then client-level values:

$$\text{DSA}_{\text{pool}} = \frac{1}{K} \sum_{k=1}^K \frac{1}{|\mathcal{O}_k^{\text{valid}}|} \sum_{o \in \mathcal{O}_k^{\text{valid}}} \text{DSA}_{k,o}.$$

DSA lies on a probability scale: a value of 0.12 means roughly a 12-percentage-point directional difference in predictive mass, not a 0.12-point accuracy difference.

DSA asks a specific question: for the same semantic source, does applying operator o move probability toward the classes associated with o during this client's training, relative to other operators? Under the stated assumptions, this is a binding-direction operator-response contrast rather than a reconstruction of the model's full internal causal mechanism.

4.2 Paired Cancellation and Binding Specificity

The following is an identification decomposition of the paired estimand, not a generative assumption about the neural network. Write

$$m_{k,o}(z, o') = s_{k,o}(z) + r_{k,o}(z, o'),$$

where $s_{k,o}(z)$ is an operator-invariant source term and $r_{k,o}$ is an operator-dependent response. Then

$$d_{k,o}(z) = r_{k,o}(z, o) - \frac{1}{|\mathcal{O}| - 1} \sum_{o' \neq o} r_{k,o}(z, o'),$$

so the operator-invariant source term cancels exactly. The decomposition therefore isolates the operator-dependent part of the paired response along the pre-registered binding direction. Semantic validity of the transformations and relevance beyond the evaluated distribution remain separate assumptions.

To test binding specificity, we preserve the number of classes assigned to each operator but randomly permute the binding and recompute DSA on the same prediction cache. Under the primary strong-CLE HFL checkpoint, true-binding DSA is 0.119644, whereas the shuffled-binding null has $p_{95} = 0.029559$ with permutation $p = 0.000999$. The gap between the true binding and this null is evidence that the measured response follows the training association rather than only a generic operator-to-class preference.

4.3 Why Consistency and Proxy Scores Are Insufficient

Within-operator consistency and cross-operator directional invariance are different objects. If multiple predictive views around the same corrupted observation are identical, their JSD is zero; predictions can nevertheless differ systematically across corruption operators. In a fixed-cache construction, within-operator JSD is zero while DSA remains 0.119644. Thus,

$$\begin{array}{l} \text{within-operator consistency} \\ \not\Rightarrow \quad \text{cross-operator directional invariance.} \end{array}$$

JSD and DSA therefore answer different questions: local consistency around a corrupted view versus cross-operator alignment with the training binding.

The same separation motivates a proxy non-identifiability result. Let Z be a proxy environment predicted by PEW and E the latent true corruption environment. Define

$$\mathcal{D}(A, B) = TV(P_{A,B}, P_A P_B).$$

With no constraint on the error channel $P(Z | E, Y)$, a fixed observable $P(Y, Z)$ can correspond to one latent world with $\mathcal{D}(Y, E) = 0$ and another with

$$\mathcal{D}(Y, E) = 1 - \sum_e P(E = e)^2 > 0.$$

Therefore $P(Y, Z)$ alone does not identify true environment dependence. A conditional bridge remains valid under sample-wise coupling:

$$\mathcal{D}_Q(Y, E) \leq \mathcal{D}_Q(Y, Z) + 2\epsilon_Q, \quad \epsilon_Q = P_Q(Z \neq E).$$

In the current BER effective distribution, the PEW family error is 0.507–0.575, so the bound becomes trivial after truncation at 1.0. Proxy statistics alone are therefore non-informative for certifying true CLE dependence in

this setting, and we evaluate mitigation with paired DSA. Appendix A gives the constructive proof and algebraic checks.

4.4 Statistical Unit

The independent evaluation unit is the **semantic source**. Multiple operator views of one source share content, so we first form source-level contrasts and then use a source-clustered bootstrap. These intervals quantify evaluation-source uncertainty conditional on the trained checkpoint; training-seed, partition, binding-map, model-selection, and cross-dataset uncertainty require separate analyses.

5 Local-versus-Communication Formation Analysis

We compare four matched arms under CLE-v2: HFL with $\gamma_{\text{CLE}} = 0$, HFL with $\gamma_{\text{CLE}} = 0.9$, Local with $\gamma_{\text{CLE}} = 0$, and Local with $\gamma_{\text{CLE}} = 0.9$. All arms share the same AugMix/JSD/DCL local baseline, use training seed 0, run for 12 rounds, and use at most 16 local batches per client per round. The mechanism factorial uses its frozen 12-round budget, distinct from the 40-round held-out objective comparison.

The matched contrasts are

$$\Delta_{\text{HFL}} = 0.119889, \quad \Delta_{\text{Local}} = 0.107960,$$

and

$$\Delta_{\text{comm}} = \Delta_{\text{HFL}} - \Delta_{\text{Local}} = 0.011929.$$

Their source-bootstrap 95% intervals are [0.118050, 0.121562], [0.106145, 0.109678], and [0.011115, 0.012712], respectively. At this evaluated seed and fixed scenario, the descriptive ratio

$$\Delta_{\text{Local}}/\Delta_{\text{HFL}} = 90.05\%$$

shows that the Local contrast is close to the HFL contrast. We call this **predominantly local-first formation**. The 90.05% ratio is descriptive rather than a per-example mediation quantity. Because the communication add-on is much smaller than the Local contrast in this matched setting, we target the local objective.

[EVIDENCE NEEDED: M2 multi-training-seed stability for the matched HFL-versus-Local four-arm factorial; training seeds 1/2 remain pending.]

6 Taxonomy-Assisted Local Mitigation

6.1 Public Environment Witness

The **Public Environment Witness (PEW)** is trained only on public carrier images with programmatically gen-

Table 1: CLE formation and local-versus-communication attribution. Numeric rows are the completed training-seed-0 factorial; M2 training seeds 1/2 remain pending.

Arm	Scope	γ_{CLE}	Operator-grid Acc. (%)	Pooled DSA	Training-seed status
HFL control	HFL	0.0	24.9300	-0.000245	seed 0 complete; 1/2 pending
HFL strong CLE	HFL	0.9	21.4367	0.119644	seed 0 complete; 1/2 pending
Local control	Local	0.0	25.5683	-0.001914	seed 0 complete; 1/2 pending
Local strong CLE	Local	0.9	22.4233	0.106046	seed 0 complete; 1/2 pending

[EVIDENCE NEEDED: M2 Formal training seeds 1/2; report seed-level uncertainty separately from source-clustered evaluation intervals.]

erated corruption supervision. It is frozen as a **coarse family-level witness** over

{clean, noise, blur, weather, digital, unknown}.

For each private fit sample, PEW produces a hard pseudo-environment $\hat{e}_i$. Its semantics come from a pre-defined corruption-family taxonomy learned from public synthetic corruptions, while private true operator and family metadata remain unavailable at training time. PEW is therefore **taxonomy-assisted**; architecture and optimizer details appear in Appendix B because the witness supplies side information rather than serving as the architectural contribution.

The two levels of granularity serve different roles: CLE-v2 construction and DSA evaluation are operator-level, whereas PEW and BER use coarse family-level pseudo-environments. Appendix C uses true operator metadata only in a non-deployable Oracle audit; the deployable witness remains family-level.

6.2 Balanced Environment Risk

For client k , task class c , and pseudo-environment e , define

$$n_{k,c,e} = \#\{i : y_i = c, \hat{e}_i = e\}.$$

$$R_{k,c,e} = \frac{1}{n_{k,c,e}} \sum_{i: y_i=c, \hat{e}_i=e} \ell_i.$$

BER assigns class-conditional environment weight

$$a_{k,c,e}^{(\gamma_{\text{BER}})} = \frac{\mathbf{1}[n_{k,c,e} \geq m] \min(n_{k,c,e}, K_{\text{cap}})^{\gamma_{\text{BER}}}}{\sum_{e'} \mathbf{1}[n_{k,c,e'} \geq m] \min(n_{k,c,e'}, K_{\text{cap}})^{\gamma_{\text{BER}}}},$$

with the frozen Formal configuration

$$\gamma_{\text{BER}} = 0.5, \quad K_{\text{cap}} = 32, \quad m = 2.$$

The client task risk is

$$R_{\text{BER},k} = \frac{1}{|\mathcal{C}_k^{\text{valid}}|} \sum_{c \in \mathcal{C}_k^{\text{valid}}} \sum_e a_{k,c,e}^{(\gamma_{\text{BER}})} R_{k,c,e}.$$

BER differs from worst-group optimization. Within each class, it sublinearly compresses the support-induced

risk-mass advantage of large pseudo-environment groups rather than equalizing group weights. Equivalently, BER performs empirical risk minimization under effective sample mass

$$q_i = \frac{a_{k,y_i,\hat{e}_i}^{(\gamma_{\text{BER}})}}{|\mathcal{C}_k^{\text{valid}}| n_{k,y_i,\hat{e}_i}}.$$

For two valid environments within a class,

$$\frac{Q_\gamma(e_1 | c)}{Q_\gamma(e_2 | c)} = \left(\frac{\min(n_{c,e_1}, K_{\text{cap}})}{\min(n_{c,e_2}, K_{\text{cap}})} \right)^{\gamma_{\text{BER}}}.$$

Hence $0 \leq \gamma_{\text{BER}} < 1$ compresses the class-conditional log-support advantage. At the current configuration, the maximum effective mass ratio between valid pseudo-environment groups is 4. A fixed-data audit verifies that the implementation matches this effective-distribution formula; Appendix B reports the audit. These properties describe the training distribution induced by BER, while trained-model behavior is evaluated separately with DSA and task utility.

7 Experiments

7.1 Protocol and Metrics

The experiments answer four questions: whether CLE creates binding-aligned behavior, whether that behavior is predominantly local-first, whether PEW+BER differs from generic ERM/group/tail-risk objectives, and whether DSA suppression persists when we change the binding map or communication base. We report DSA together with operator-grid accuracy and frozen task metrics: client-average accuracy (Avg), minimum client accuracy (Worst), worst valid class-operator-cell accuracy (WCCA), and the mean within-class accuracy range across operators (CFG; lower is better).

The experiments use matched contrasts rather than a single cross-protocol leaderboard. The held-out map2 study is the main comparison of local risk objectives; map1 changes the class-operator binding while keeping the remaining controlled factors fixed; FedDF-fidelity changes the HFL communication base. Table 2 summarizes the information available during training and evaluation.

Table 2: Information boundary for completed and frozen comparison protocols. Operator identities and binding maps are sealed from deployable training.

Protocol role	Private task	Train-time environment information	Communication context	Evaluation-only information
CLE formation / local-first	CIFAR-10	none	Local or strict AsymHFL	operator, binding, paired grid
Axis I map2 objectives	CIFAR-10	none or PEW family proxy, by arm	strict AsymHFL	operator, binding, paired grid
Cross-binding map1	CIFAR-10	PEW family proxy for PEW+BER	strict AsymHFL	operator, binding, paired grid
Second communication base	CIFAR-10	PEW family proxy for PEW+BER	FedDF-fidelity	operator, binding, paired grid
Axis II S2-v2 context	CIFAR-10	method-specific; no BER	protocol-matched HFL family	operator, binding, paired grid

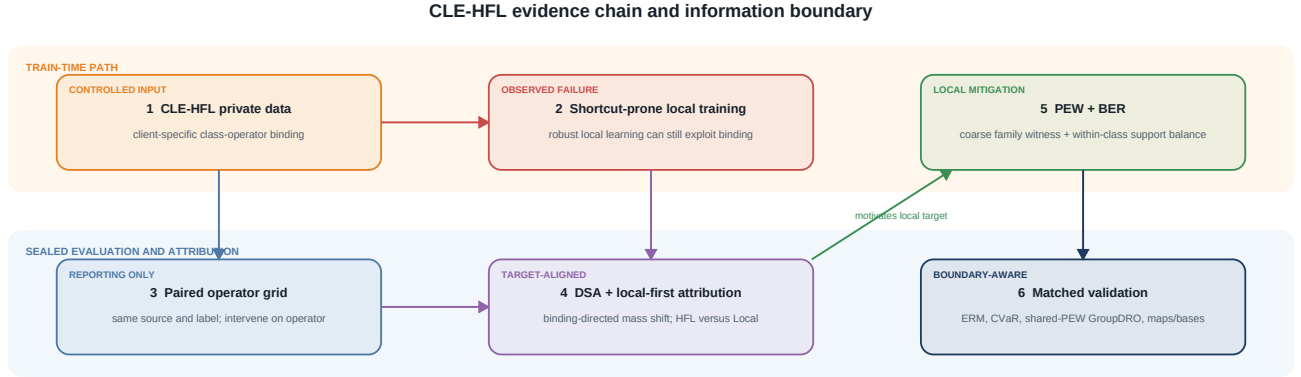


Figure 1: CLE-HFL evidence chain and information boundary. Training-time PEW+BER uses a coarse family proxy. Operator identities and the binding map are sealed from training and used only for controlled construction or post-training DSA evaluation.

7.2 Axis I: Matched Shortcut-Objective Attribution

A five-arm 12-round screen was used for method selection only. JTT therefore remains screening-only and is not part of the Formal comparison. The held-out map2 Formal uses 40 rounds and four matched arms: ERM, protocol-matched CVaR-DRO, PEW+GroupDRO, and PEW+BER. They share the same partition, training/evaluation seed, initial states, private batch trajectory, and strict HFL communication protocol. PEW+GroupDRO and PEW+BER receive the same PEW group assignments.

For training seed 0, the source-paired DSA differences are

$$\text{DSA}(\text{ERM}) - \text{DSA}(\text{BER}) = 0.182567,$$

with 95% interval $[0.180596, 0.184407]$,

$$\text{DSA}(\text{GroupDRO}) - \text{DSA}(\text{BER}) = 0.136834,$$

with $[0.134952, 0.138706]$, and

$$\text{DSA}(\text{BER}) - \text{DSA}(\text{CVaR}) = -0.011240,$$

with $[-0.012355, -0.010092]$. These intervals resample semantic sources conditional on the seed-0 checkpoints.

Across training seeds 0/1/2, $\text{DSA}(\text{ERM}) - \text{DSA}(\text{BER})$ is 0.179060 ± 0.008594 , and $\text{DSA}(\text{GroupDRO}) - \text{DSA}(\text{BER})$

is 0.133947 ± 0.011518 ; both contrasts favor BER in all three seeds. $\text{DSA}(\text{BER}) - \text{DSA}(\text{CVaR})$ is -0.015184 ± 0.015411 , while BER improves operator-grid accuracy over CVaR by 5.9567 ± 2.3151 percentage points. The pre-registered multi-seed gates pass, yielding `GO_MAP2_TRAINING_SEED_STABILITY`.

The shared-PEW comparison separates access to PEW groups from the BER objective: PEW+GroupDRO retains pooled DSA 0.2054 ± 0.0080 , whereas PEW+BER reaches 0.0715 ± 0.0113 . CVaR-DRO is closer at 0.0867 ± 0.0055 , but BER also has higher pooled operator-grid accuracy, Avg, Worst, and WCCA, and lower CFG. Per-client DSA is not uniformly ordered: CVaR-DRO is lower on c0 and c1, while BER is lower on c2 and c3. Among these four matched objectives, PEW+BER therefore gives the best pooled shortcut-utility trade-off on held-out map2 without per-client DSA dominance over CVaR-DRO.

7.3 Shared Validation: Binding Map and Communication Base

Table 4 summarizes completed PEW+BER comparisons on the original binding map, map1, and FedDF-fidelity. A second private dataset remains a separate evidence slot.

On map1, only the client-specific class-operator binding changes; the partition, evaluation seed, training seed, initial weights, public data, evaluation grid, and frozen PEW remain fixed. Pooled DSA changes from 0.113761 to 0.049531 , a 56.46% reduction, and all four clients move in

Table 3: Held-out map2 matched objective comparison over training seeds 0/1/2 (mean $\pm$ sample standard deviation).

Arm	DSA $\downarrow$	Operator-grid Acc. $\uparrow$	Last-10 Avg $\uparrow$	Last-10 Worst $\uparrow$	WCCA $\uparrow$	CFG $\downarrow$
ERM	0.2505 $\pm$ 0.0087	20.4950 $\pm$ 0.3718	19.9589 $\pm$ 0.1212	16.3780 $\pm$ 0.1259	0.0000 $\pm$ 0.0000	32.7467 $\pm$ 0.1708
CVaR-DRO	0.0867 $\pm$ 0.0055	19.2206 $\pm$ 1.1055	18.5788 $\pm$ 0.3102	14.2636 $\pm$ 0.2640	0.1167 $\pm$ 0.0629	27.6750 $\pm$ 0.1885
JTT Formal	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>
PEW+GroupDRO	0.2054 $\pm$ 0.0080	21.4517 $\pm$ 0.5783	20.9963 $\pm$ 0.0924	17.5309 $\pm$ 0.5148	0.0333 $\pm$ 0.0144	31.8950 $\pm$ 0.9747
PEW+BER	0.0715 $\pm$ 0.0113	25.1772 $\pm$ 1.2522	24.5466 $\pm$ 0.1098	20.7516 $\pm$ 0.6126	2.6500 $\pm$ 0.0750	22.7825 $\pm$ 0.5428

JTT remains screening-only; an O1 Formal row should be populated only if a formal JTT comparison is later authorized and completed.

Table 4: Within-setting PEW+BER comparisons across binding maps and communication bases.

Private task	Binding / protocol	HFL base	Base DSA	PEW+BER DSA	Relative reduction	Avg delta
CIFAR-10	original map	Asymmetric HFL	0.119644	0.041252	65.52%	+1.7323
CIFAR-10	map1	Asymmetric HFL	0.113761	0.049531	56.46%	+0.7040
CIFAR-10	held-out map2 (3 seeds)	Asymmetric HFL	0.250530	0.071470	71.47%	+4.5877
CIFAR-10	original map	FedDF-fidelity	0.136989	0.029012	78.82%	-0.6660
Second private task	dataset/protocol pending	pending	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

[EVIDENCE NEEDED: second private dataset; dataset and protocol pending.]

the same direction. Operator-grid accuracy changes from 20.9450% to 21.4583%. This reproduces DSA suppression on a second controlled binding direction at seed 0.

The second communication-base experiment uses native single-view CE with a protocol-matched **FedDF-fidelity** adapter. Base and BER arms share initialization, private batches, public distillation, and training budget. Pooled DSA changes from 0.136989 to 0.029012, while operator-grid accuracy changes by -0.0400 points and last-five Avg by -0.6660 points. The frozen utility gate fails, and both arms fall below the pre-registered 20% operator-grid learning floor. We use this experiment as supporting evidence that DSA suppression also appears under a second protocol-matched communication base; it does not support a cross-base utility gain.

[EVIDENCE NEEDED: second private dataset; dataset and protocol pending.]

7.4 Axis II: HFL Mechanism-Family Context

Axis II asks a different question from the matched objective study: whether representative heterogeneous-FL communication and robustness mechanisms naturally remove CLE. The planned context includes Local/ERM, FedMD, FedProto, FedTGP, FedDF, KT-pFL, FCCL, RHFL, AugHFL, and RAHFL under a protocol-matched S2-v2 budget. These implementations provide mechanism-family context rather than official-recipe leaderboards; FedTGP, for example, is a protocol-matched core adapter. Existing benchmark and smoke outputs are engineering checks and are not used as scientific results.

[EVIDENCE NEEDED: S2-v2 Formal scientific metrics for the HFL mechanism-family context table.]

7.5 Environment Correspondence and Granularity

A non-deployable Oracle audit compares true family grouping, true operator grouping, and a random operator correspondence. Random operator DSA exceeds Oracle operator DSA by 0.052112, with source-bootstrap 95% interval $[0.051231, 0.052983]$, showing that the environment correspondence matters in this audit. The Oracle family-operator DSA gap is 0.001449, with $[0.001086, 0.001790]$. The small granularity gap is specific to this controlled audit and provides no empirical reason to introduce an operator-level PEW in the current paper. Appendix C reports the complete Oracle table.

7.6 Remaining Evidence Placeholders

The internal manuscript keeps unresolved evidence visible. M1, the three-seed map2 objective comparison, is complete. M2 remains the multi-seed extension of the local-first factorial. M3 is the second-private-dataset slot, with both dataset and protocol pending. S1 is a bounded taxonomy-stress slot with protocol pending. S2-v2 has engineering benchmark/pairing work but no Formal scientific metrics in the current evidence package. O1 remains an optional JTT Formal only if the final paper makes an empirical JTT comparison.

[EVIDENCE NEEDED: M2 local-first training seeds 1/2.]

[EVIDENCE NEEDED: bounded taxonomy stress test; protocol pending.]

8 Discussion and Limitations

Shortcut suppression is not the same as task utility. On the original Asymmetric-HFL setting, PEW+BER

Table 5: Protocol-matched HFL mechanism-family context (Axis II). Scientific metric cells remain pending until S2-v2 Formal results are completed and audited.

Method	Model heterogeneity	Local robustness	Communication	DSA ↓	Operator Acc. ↑	Avg ↑
Local/ERM	yes	none	none	<i>pending</i>	<i>pending</i>	<i>pending</i>
FedMD adapter	yes	none	public-logit exchange	<i>pending</i>	<i>pending</i>	<i>pending</i>
FedProto adapter	yes	none	class prototypes	<i>pending</i>	<i>pending</i>	<i>pending</i>
FedTGP core adapter	yes	none	trainable global prototypes	<i>pending</i>	<i>pending</i>	<i>pending</i>
FedDF-fidelity	yes	none	public distillation	<i>pending</i>	<i>pending</i>	<i>pending</i>
KT-pFL-fidelity	yes	none	knowledge transfer	<i>pending</i>	<i>pending</i>	<i>pending</i>
FCCL protocol adapter	yes	none	cross-correlation collaboration	<i>pending</i>	<i>pending</i>	<i>pending</i>
RHFL adapter	yes	SCE / label-noise robustness	confidence-weighted collaboration	<i>pending</i>	<i>pending</i>	<i>pending</i>
AugHFL-fidelity	yes	AugMix/JSD	heterogeneous collaboration	<i>pending</i>	<i>pending</i>	<i>pending</i>
RAHFL anchor	yes	AugMix/JSD/DCL	asymmetric collaboration	<i>pending</i>	<i>pending</i>	<i>pending</i>

Existing smoke/benchmark outputs are engineering checks only; this table requires S2-v2 Formal scientific metrics.

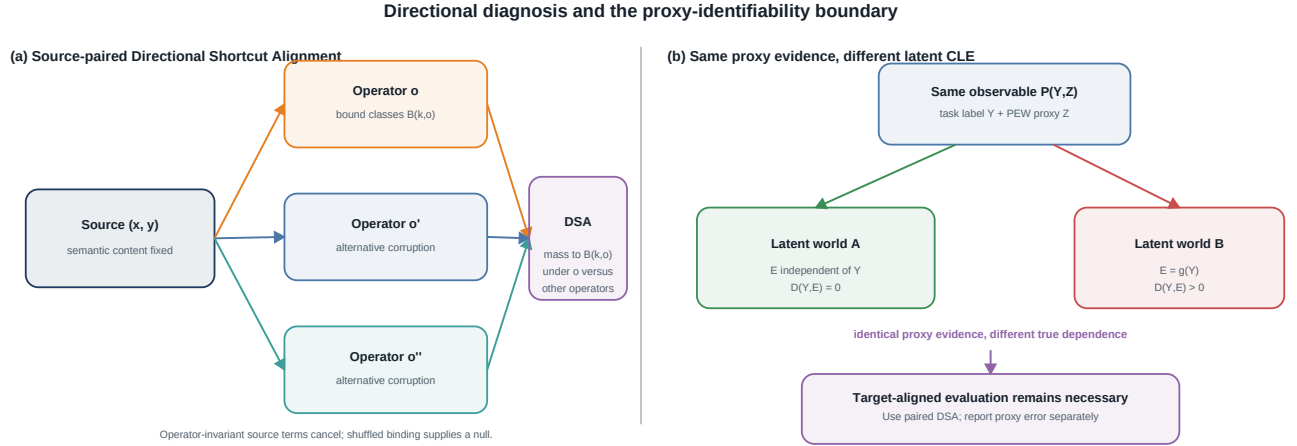


Figure 2: Paired directional diagnosis and proxy-identifiability boundary. DSA compares bound-class probability mass across operators for the same source. Identical observable label-proxy evidence can correspond to latent worlds with different true dependence.

reduces pooled DSA by 65.52%, but not every client-level utility metric improves. Under FedDF-fidelity, DSA falls from 0.136989 to 0.029012, while the frozen utility gate fails. On held-out map2, BER has the best pooled shortcut-utility trade-off across the four matched objectives, yet CVaR-DRO has lower DSA on c0 and c1. We therefore report DSA and task utility as separate outcomes.

PEW is taxonomy-assisted. It learns a fixed coarse corruption-family taxonomy from public synthetic corruptions, unlike taxonomy-free environment inference such as EIL [1]. BER has an exact effective-distribution interpretation for the resulting proxy groups, but the current PEW error makes the conditional bridge to the true environment trivial. The proxy non-identifiability result formalizes this gap. We therefore judge behavioral mitigation with paired DSA rather than proxy statistics alone. The deployable method is family-level; true operator information is used only for controlled construction, sealed DSA evaluation, and Oracle audits.

The evidence is still narrow in training and data variation. The held-out map2 objective comparison spans training seeds 0/1/2, while the local-first factorial, map1 replication, and FedDF-fidelity experiment use seed

0. All core experiments use one controlled CIFAR-10 private task and a fixed non-IID partition. Source-clustered bootstrap covers evaluation-source variation conditional on each checkpoint; it is not training-seed or dataset-level uncertainty. **[EVIDENCE NEEDED: second private dataset; dataset and protocol pending.]** **[EVIDENCE NEEDED: bounded taxonomy stress test; protocol pending.]** Client identity also couples architecture with a particular non-IID data slice, so the current per-client differences cannot isolate architecture capacity.

9 Conclusion

We study class-corruption entanglement in model-heterogeneous federated learning, where client-specific class-operator associations can make corruption a directional class cue. Source-paired operator interventions and Directional Shortcut Alignment measure this binding-directed response under explicit identification assumptions, while proxy non-identifiability shows why pseudo-environment improvements alone cannot certify reduced CLE dependence. The seed-0 factorial compar-

ing HFL and Local training finds a predominantly local-first pattern, motivating a taxonomy-assisted intervention based on coarse family-level PEW and BER. On held-out map2, the four-arm comparison reproduces across training seeds 0/1/2: PEW+BER has lower pooled DSA and higher pooled task utility than ERM, CVaR-DRO, and PEW+GroupDRO. Seed-0 experiments also reproduce DSA suppression on another binding map and under FedDF-fidelity, where the utility gate fails. The remaining evidence gaps concern local-first training-seed stability, a second private dataset, and bounded taxonomy mismatch.

A DSA Properties and Proxy Non-identifiability

A.1 Exchangeable Zero and Affine Property

If predictions for the same source are exchangeable across all operators,

$$p(\cdot \mid T_o(x)) = p(\cdot \mid T_{o'}(x)), \quad \forall o, o',$$

then every source-level DSA contrast is zero. The fixed-cache exchangeable projection yields -2.50×10^{-20} ; the largest empirical absolute DSA under the $\gamma_{\text{CLE}} = 0$ controls is 0.001914.

DSA is affine in predictive probabilities. For

$$p_\lambda = (1 - \lambda)p_0 + \lambda p_1,$$

$$\text{DSA}(p_\lambda) = (1 - \lambda) \text{DSA}(p_0) + \lambda \text{DSA}(p_1).$$

The following are numerical identity checks rather than statistical precision claims. An 11-point fixed-cache validation has maximum numerical error 2.78×10^{-17} . In a controlled probability-simplex injection, DSA recovers a known binding-aligned response with maximum numerical error 6.66×10^{-16} . Adding an identical displacement to every operator view of the same source changes DSA by only 8.33×10^{-17} , numerically validating paired cancellation.

A.2 Constructive Proxy Non-identifiability

Fix an observable $P(Y, Z)$, with at least two positive-probability values of Y , and leave $P(Z \mid E, Y)$ unconstrained. One latent world can choose E independent of (Y, Z) , yielding $\mathcal{D}(Y, E) = 0$. Another can set $E = g(Y)$ for a nonconstant mapping while retaining the same $P(Z \mid Y)$, yielding

$$\mathcal{D}(Y, E) = 1 - \sum_e P(E = e)^2.$$

The two worlds have identical observable $P(Y, Z)$ but different true dependence. In the balanced ten-class, five-environment cache construction, these values are 0 and 0.8. The construction proves non-identifiability for an unconstrained proxy channel; it is a proof device rather than a model of a natural corruption-generating process.

A.3 Source-Level Inference

The main bootstrap first forms source-level directional contrasts and then resamples sources. A conservative bounded-variable calculation for $n = 1000$, $\delta = 0.05$ gives Hoeffding radius 0.085894, but the paper uses source-clustered bootstrap intervals. Neither calculation covers training randomness.

B PEW and BER Implementation/Audit Details

PEW uses public CIFAR-100 carrier images with programmatically generated coarse labels over clean, noise, blur, weather, digital, and unknown environments. Unknown is generated by sequentially composing two distinct base families. Severity ranges from 1 to 5. The frozen witness is a small CNN predicting six environments, five severity levels, and a 32-dimensional embedding, trained with environment cross-entropy plus 0.25 times severity cross-entropy for five epochs using Adam at learning rate 10^{-3} . Checkpoint and unknown-threshold selection use public validation only.

A fixed-data CPU audit verifies BER’s effective empirical distribution with maximum implementation/formula mismatch 7.32×10^{-16} . On the strict-fit audit distribution, equal-client dependence changes from $TV(Y, \hat{E}) = 0.487505$ to 0.199744 (59.03% relative drop). Reporting-only true-family dependence changes from 0.634914 to 0.433513 (31.72% relative drop), and true-family environment-only Bayes advantage changes from 0.313840 to 0.240863 (23.25% relative drop). These quantities audit BER’s effective training distribution. The deployable rule does not use true environment metadata, and trained-model behavior is evaluated separately with DSA and task accuracy.

C Additional Experimental Evidence

C.1 Original-Map PEW+BER

On the original strong-CLE map, pooled DSA changes from 0.119644 to 0.041252, an absolute reduction of 0.078392 and relative reduction of 65.52%; the source-bootstrap 95% interval for the DSA difference is $[0.077191, 0.079601]$. Per-client DSA reductions are $[0.111312, 0.011317, 0.115762, 0.075178]$. Last-five intervention-minus-baseline deltas are Avg +1.7323, Worst

−0.3760, WCCA +0.3500, and CFG −9.2600 percentage points. The shortcut-mitigation criterion passes, but the frozen full-utility criterion fails because Worst does not meet its pre-registered threshold.

C.2 Oracle Correspondence and Granularity

Table 6: Oracle correspondence and granularity audit.

Grouping	Pooled DSA	Operator-grid Avg
Oracle family	0.017232	21.9333
Oracle operator	0.015783	21.5200
Random operator	0.067895	18.7817

The corresponding last-five Avg/Worst values are (21.0960/17.8400), (20.0273/15.8827), and (18.3933/14.7867) for Oracle family, Oracle operator, and Random operator, respectively.

Oracle labels are unavailable to the deployable method. The Random-minus-Oracle-operator DSA difference is 0.052112, 95% interval [0.051231, 0.052983]. The Oracle-family-minus-Oracle-operator difference is 0.001449, [0.001086, 0.001790]. In this audit, the family-operator DSA gap is small relative to the random-correspondence gap. The deployable PEW therefore remains coarse family-level in the current paper.

C.3 Screening-Only JTT Boundary

JTT was evaluated only in the 12-round screening stage used for method selection. Those screening values are used for design history only. The main text discusses JTT conceptually [11]; a formal JTT comparison remains pending unless O1 is separately authorized and completed.

D Submission Evidence Table Skeletons

The following draft-only tables keep unresolved evidence slots visible in the internal manuscript. Red pending cells are placeholders, not results. Set `\draftskeletontofalse` in `main.tex` to hide fully pending tables for an evidence-only compilation.

D.1 Bounded Taxonomy Stress

D.2 PEW and BER Distribution Audit

D.3 Per-Client Architecture Audit

D.4 Uncertainty Separation

D.5 Computation and Communication Cost

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Table 7: Placeholder for a bounded taxonomy-stress experiment. The stress definition and evaluation protocol remain pending.

Method	PEW proxy quality	Pooled DSA ↓	Stress-specific DSA ↓	Avg ↑
ERM	—	<i>pending</i>	<i>pending</i>	<i>pending</i>
CVaR-DRO	—	<i>pending</i>	<i>pending</i>	<i>pending</i>
PEW+BER	<i>pending</i>	<i>pending</i>	<i>pending</i>	<i>pending</i>

[EVIDENCE NEEDED: bounded taxonomy stress test; protocol pending.]

Table 8: PEW/BER effective-distribution audit on the completed strict-fit cache. These are mechanism audits, not trained-model performance guarantees.

Audit quantity	Before BER	After BER
$TV(Y, \hat{E})$	0.487505	0.199744
$TV(Y, E_{\text{family}})$	0.634914	0.433513
Environment-only Bayes advantage	0.313840	0.240863

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Table 9: Per-client held-out map2 DSA over training seeds 0/1/2 (mean $\pm$ sample standard deviation). Client differences cannot be causally attributed to architecture because architecture and non-IID data slice are paired.

Client	Architecture	ERM DSA	CVaR DSA	GroupDRO DSA	PEW+BER DSA
c0	ResNet10	0.1943 ± 0.0061	0.0408 ± 0.0066	0.1309 ± 0.0249	0.0532 ± 0.0043
c1	ResNet12	0.2447 ± 0.0192	0.0672 ± 0.0042	0.1762 ± 0.0138	0.0703 ± 0.0084
c2	ShuffleNet	0.2776 ± 0.0080	0.1277 ± 0.0200	0.2557 ± 0.0085	0.1130 ± 0.0307
c3	MobileNetV2	0.2856 ± 0.0104	0.1109 ± 0.0115	0.2589 ± 0.0065	0.0494 ± 0.0181

Table 10: Separation of training-seed and evaluation-source uncertainty.

Experiment	Training seeds	Seed-level summary	Source-level interval
M1 map2 objectives	0/1/2	GO; ERM-BER 0.1791 ± 0.0086	clustered bootstrap per checkpoint
M2 local-first	0 complete; 1/2 pending	<i>pending</i>	clustered bootstrap per checkpoint
Second private task	dataset/protocol pending	<i>pending</i>	pending

[EVIDENCE NEEDED: M2 and second-private-task seed summaries; keep seed-level variation separate from checkpoint-conditional source-bootstrap intervals.]

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Table 11: Planned computation and communication audit. PEW is a one-time public pretraining cost; BER does not alter the communication protocol.

Method	Training stages	Extra public model	GPU time	Communication change	Status
ERM	1	no	<i>pending</i>	none	pending benchmark accounting
CVaR-DRO	1	no	<i>pending</i>	none	pending benchmark accounting
JTT	2	no	<i>pending</i>	none	O1 pending
PEW+GroupDRO	1 + frozen PEW	yes	<i>pending</i>	none	pending accounting
PEW+BER	1 + frozen PEW	yes	<i>pending</i>	none	pending accounting

[EVIDENCE NEEDED: Populate only from benchmark/Formal timing manifests; do not estimate missing runtimes.]